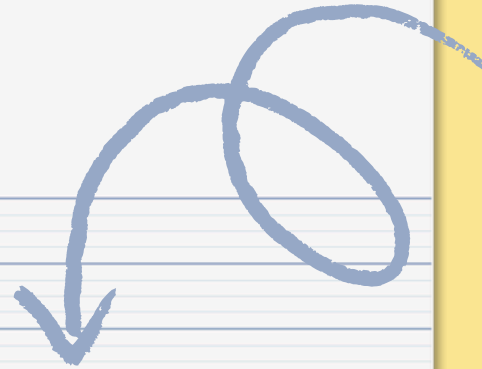
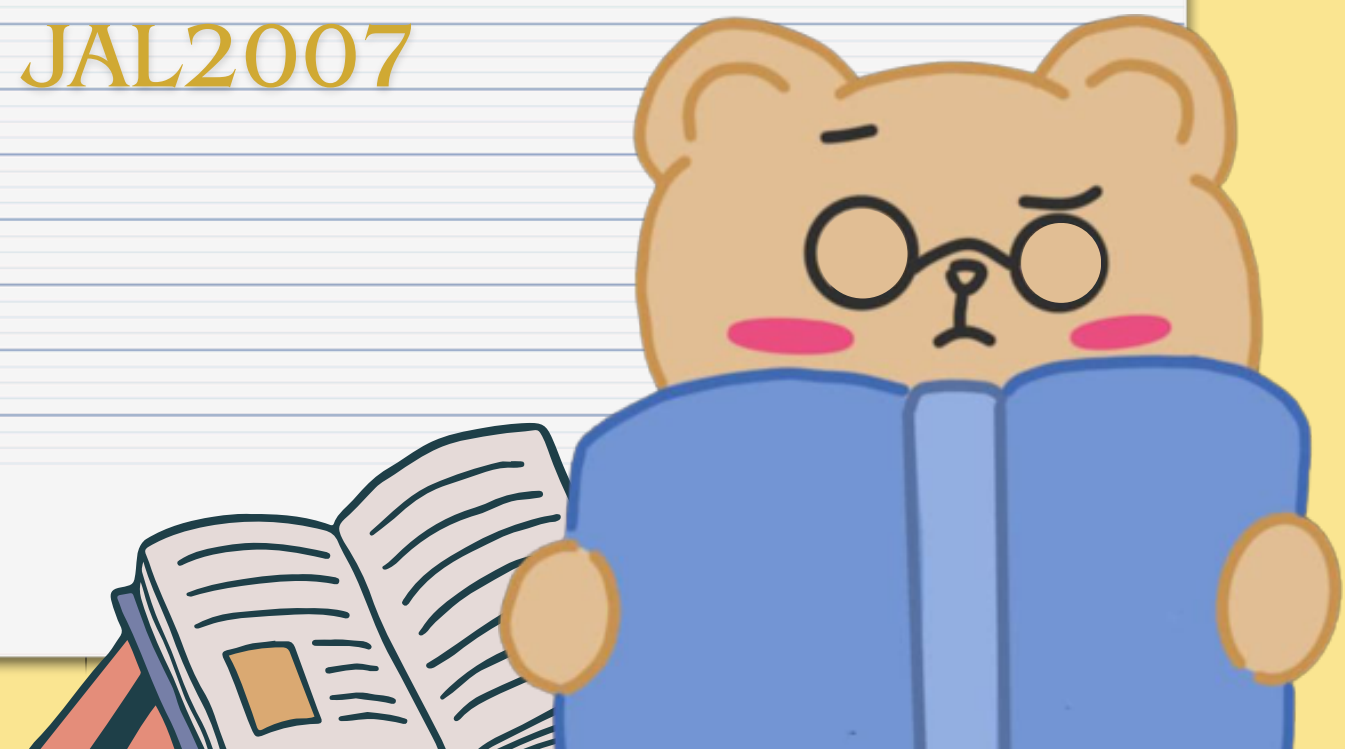
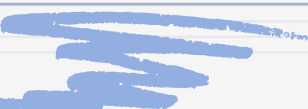
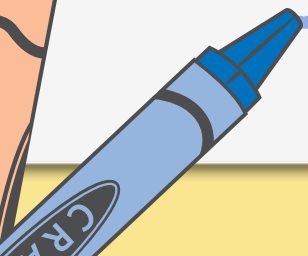
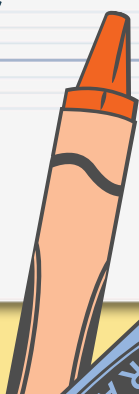




LINK AGGREGATION AND ZONING

STORAGE TECHNOLOGIES JAL2007

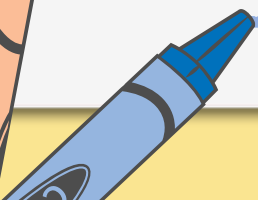
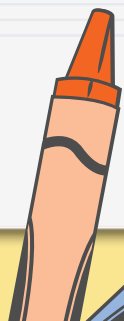


what is link aggregation?

- Combines multiple links into one logical link.
- Increases bandwidth and reliability.
- If one link fails, others continue working.

what is zoning?

- Divides a network into logical zones.
- Controls which devices can communicate.
- Improves security and performance.



Link Aggregation

Link Aggregation is the process of combining two or more physical network connections to form a single logical connection.

It is also commonly called:

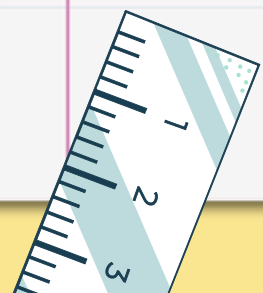
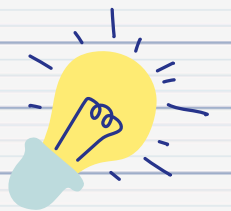
- LAG - Link Aggregation Group
- Port Trunking
- EtherChannel - commonly used terminology in Cisco networks

Example: Link Aggregation

A server uses 4×1 Gbps links.

$1 \text{ Gbps} + 1 \text{ Gbps} + 1 \text{ Gbps} + 1 \text{ Gbps} = \text{Up to } 4 \text{ Gbps}$

→ Multiple links work together to provide higher bandwidth and reliability.



How Link Aggregation Works

Basic Working

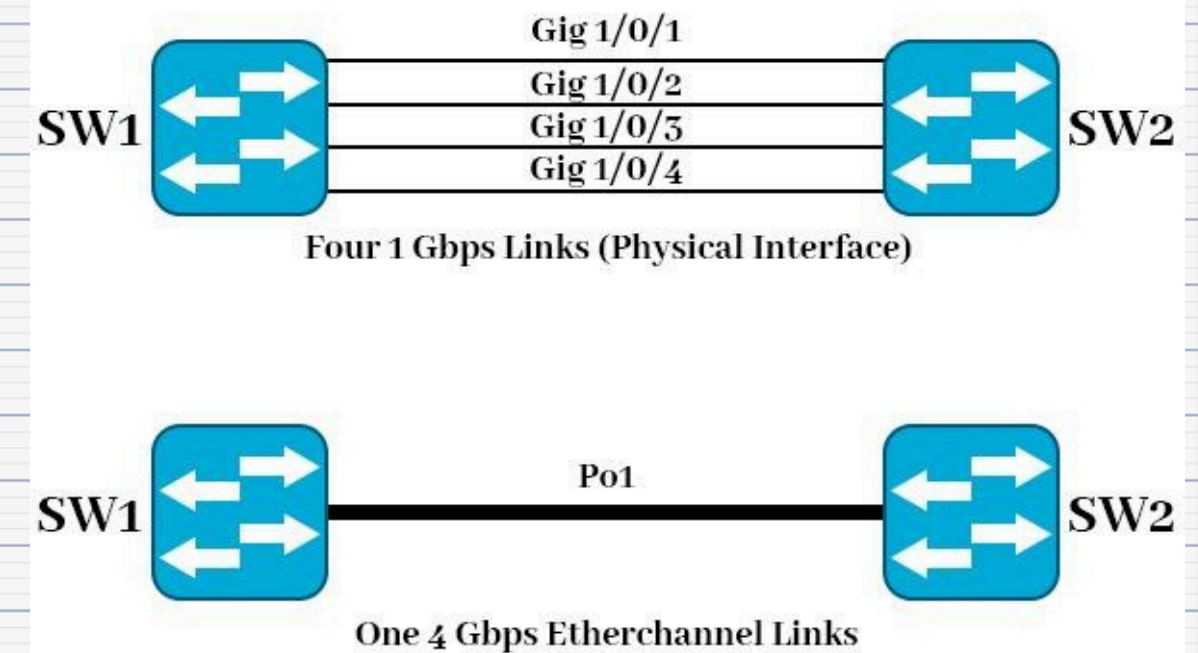
- Connect multiple cables between two devices.
- Combine them into one logical link.
- Traffic is shared across the links.
- If one link fails, the others continue working.

4 Physical Links → 1 Logical Link

Protocol

LACP (Link Aggregation Control Protocol)

- IEEE 802.1AX standard.
- Helps automatically create and maintain the link group



zoning

Zoning divides a network into separate logical areas called zones.

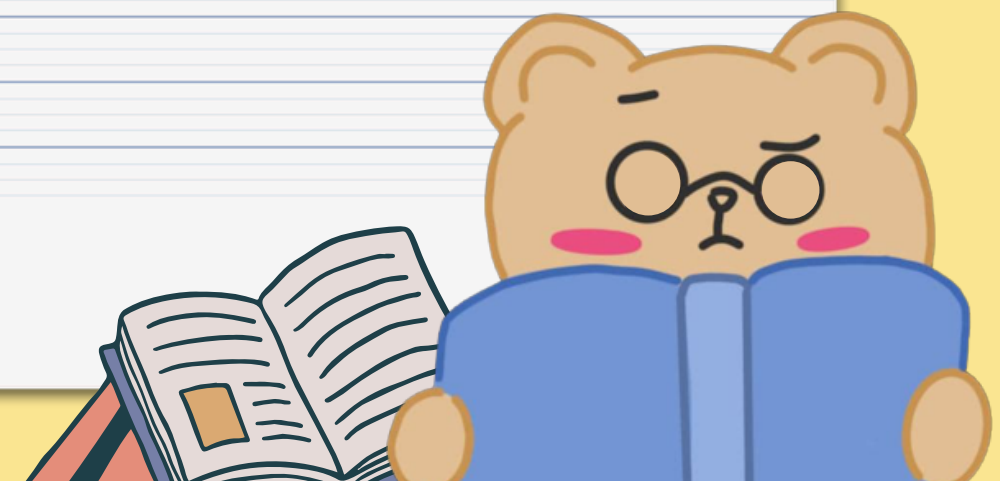
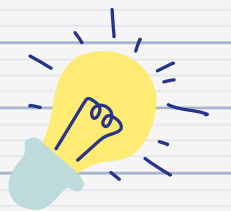
Each zone can have different:

- Security policies
- Access permissions
- Devices
- Services

Example :

A company network can be divided into:

- Employee Zone
- Server Zone
- Guest Zone



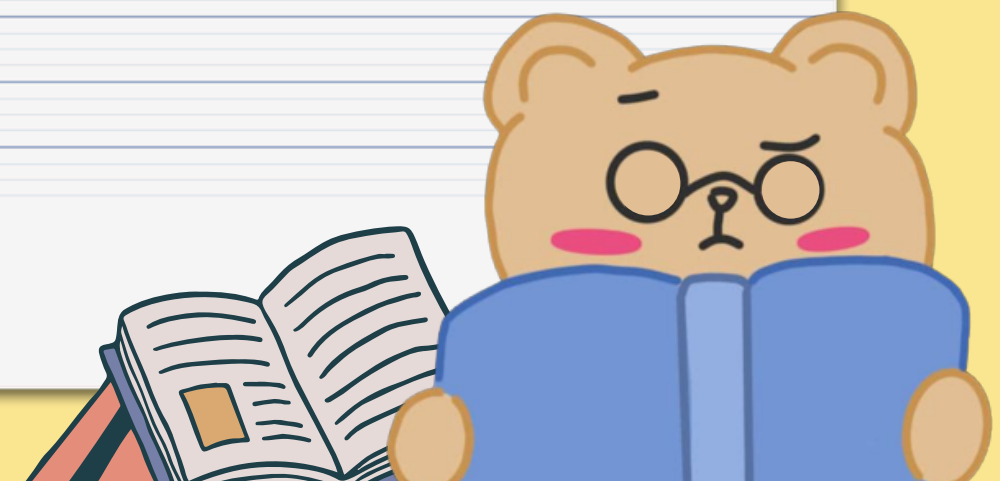
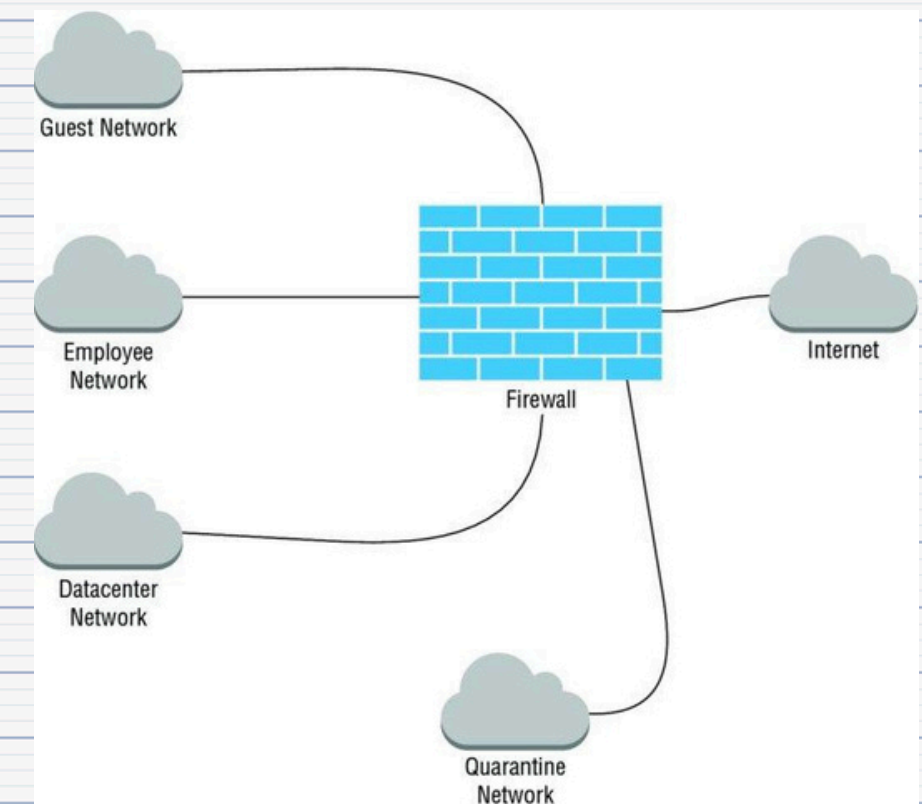
How Zoning Works

Working Process

1. Network devices are grouped into different zones.
2. Security rules are applied between zones.
3. Traffic is inspected when it moves between zones.
4. Only authorized communication is allowed.

Common Technologies

- VLANs
- Firewalls
- Access Control Lists (ACLs)
- Network segmentation
- In storage networks: SAN zoning



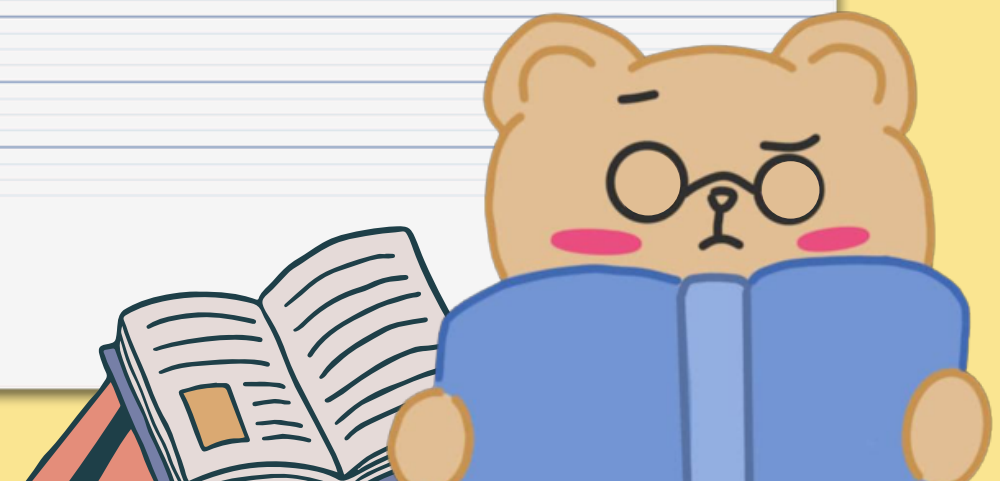
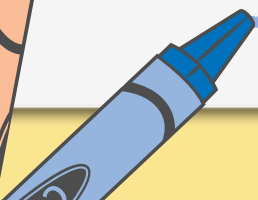
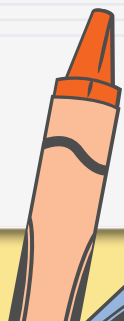
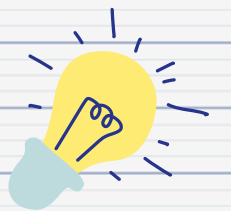
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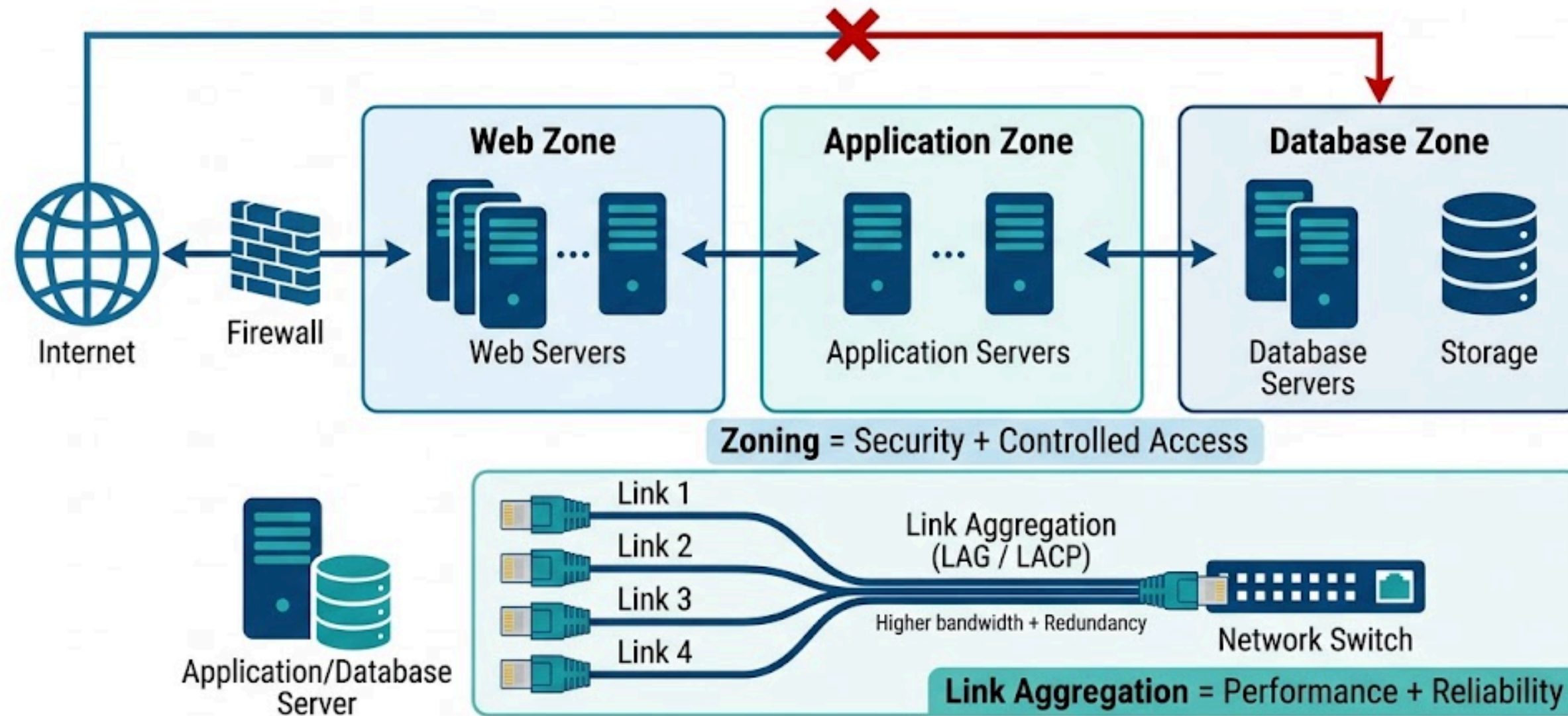
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Real-World Example: Link Aggregation + Zoning in a Data Center

A modern modern enterprise data center network combining both technologies.



Link Aggregation improves network performance and availability, while **Zoning** improves security and controls communication.

**Thank
You**

